



***From Theory to
Impact***

The Impact of Mathematical Sciences on Nation Building

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Optimization

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LET US GO BACK SMALL

They did it by hand.

No computer. No calculator. No electricity. Just paper, pen, and a stubborn mind.

Geometry

How to measure the world

Trigonometry

How to navigate it

Calculus

How things change

Differential equations

How systems behave

Probability

How to reason under uncertainty

Logarithms

How to compute the impossible

AND SEE WHAT IT BUILT

Four equations. One modern world.

James Clerk Maxwell wrote four equations describing electricity and magnetism. From those four equations came radio, television, radar, satellites, and the phone in your pocket this morning.



The aeroplane

Fluid dynamics and calculus lifted iron off the ground.



Radioactivity

Exponential decay: the mathematics of the half-life.



Global mobility

Trigonometry and orbital mechanics put GPS in the sky.



Electricity

Maxwell's equations gave us the entire electrical age.



Modern medicine

Fourier transforms are why an MRI can see inside you.



Insurance & pensions

Probability turned risk into something a nation can carry.

THEY HAD NOTHING. THEY BUILT EVERYTHING.

So what is our own excuse?

- Should our children still be dying from what we already know how to cure?
- Should our people still be hungry in a country with this much land?
- Should our grid still be collapsing in the year 2026?
- Should we still be planning a whole nation on guesswork?

The machine on your desk solves in one second what took Kepler four years.

WHERE WE ARE STANDING TODAY

The bottleneck has moved.

For all of history, mathematics was limited by the human being holding the pencil. That limit is gone.

FROM

The machine

Can we compute it?

TO

The mind

Will we apply it?

"The problem is too complex" has expired as an excuse. Today it is a choice.

LET US DEFINE OUR TERMS

What is nation building?

It is not concrete. If concrete were enough, we would already be a developed nation.

The deliberate building of the systems, institutions, identity and capabilities by which a people becomes a functioning, self-sustaining, dignified nation.

In our own language:

Moving a nation from a LOCAL maximum to the GLOBAL maximum.

A local maximum is a comfortable place to die. From where you are standing, every direction looks like a step down. So you stay there, and you call it stability.

Nation building is not a scalar.

SCALAR

Magnitude only

Plenty people.
Plenty resources.
Plenty noise.

VECTOR

Magnitude + Direction

Knowing exactly where
the mountain is.

We have magnitude in abundance. What we lack is direction.

Whose headache is it?

Government

Academia

The youth

The elders

Business

Every citizen

It is a coupled system.

Every one of us is a variable inside somebody else's equation. So the question is not whose headache it is. The question is: what is your coefficient, and what is its sign? Positive, negative, or (most commonly in this country) zero.

LET US LOOK AT OURSELVES HONESTLY

Nigeria today.

~51%

living below the international
poverty line, 2025

World Bank

~24%

inflation, early 2025
(31% average in 2024)

IMF

309

harmonised learning score
(300 = minimum, 625 = advanced)

World Bank / UNESCO

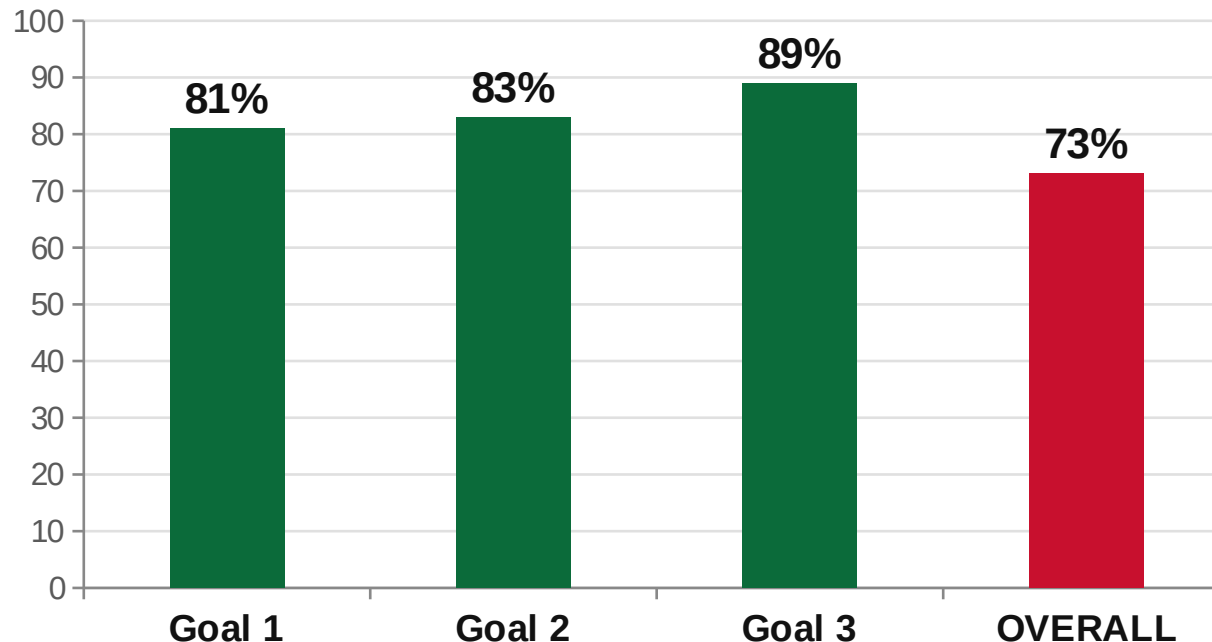
\$100bn+

national infrastructure
deficit

NBS

SOMEBODY ACTUALLY DID THE MATHEMATICS

Our national plan cannot deliver what it promises.



Multi-objective optimisation applied to Nigeria's SDG Agenda 2030: individual goals achievable at 81%, 83% and 89%, but the overall function caps at 73%.

The goals are fighting each other for the same money.

It is not that we lack will. We lack feasibility.

The structure is the body. Mathematics is the nervous system.

1

Statistics

Sampling · estimation · error bounds

2

Planning & budget

Constrained optimisation

3

Security

Graph theory · Bayesian inference

4

Justice

Queueing theory

5

Public health

Epidemiological modelling

6

Energy

Load balancing · economic dispatch

ONE DISCIPLINE UNDERNEATH ALL OF IT

Sixteen sectors. One foundation.

Security

Peace

Justice

Governance

Health

Education

Planning

Logistics

Employment

Innovation

Agriculture

Food security

Entertainment

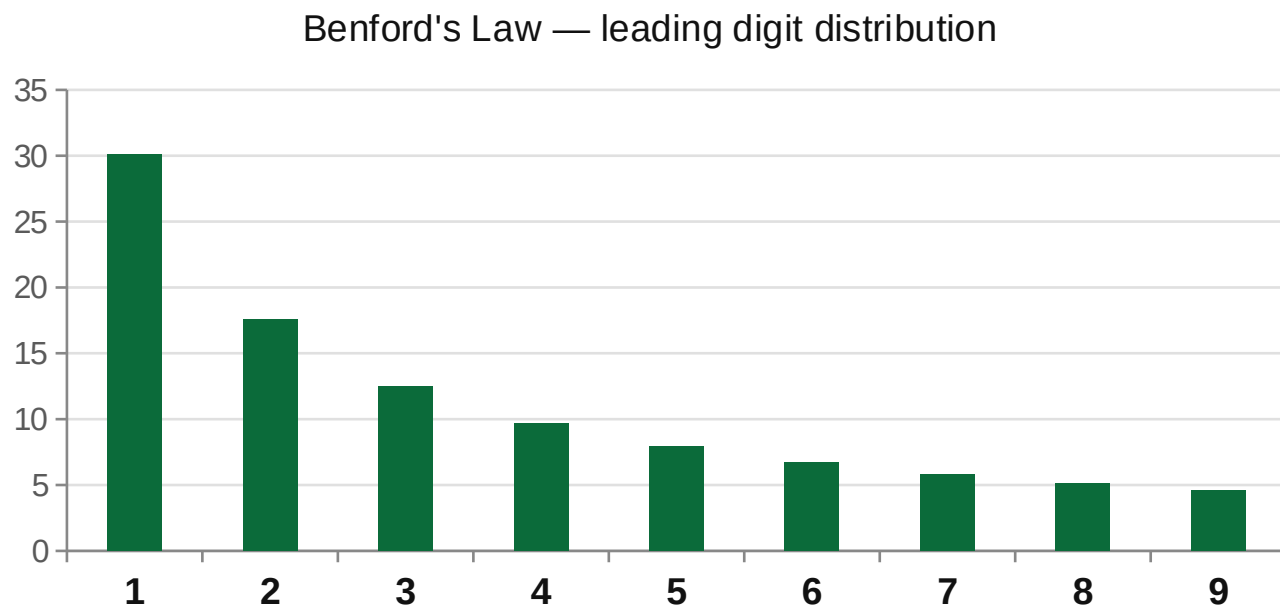
Energy

Anti-corruption

Finance

That is not coincidence. That is the definition of a foundational science.

Mathematics catches liars. It does not need a confession.



In genuine numerical data, the leading digit 1 shows up about 30% of the time.

Human beings are very, very bad at faking randomness.

An inflated contract carries a statistical fingerprint, before anybody confesses and before anybody testifies.

THE COST OF IGNORING MATHEMATICS

**is never a crash.
It is a slow bleeding.**

Ignore physics and a bridge falls down. Everybody sees it, everybody talks about it. Ignore mathematics and nothing collapses. You simply operate at 60% of what you could have been, forever, and you call it "our peculiar Nigerian challenges."

Mathematics is the only argument in the room that has no tribe.

Six things we must do.

1

Change what we reward in academic mathematics. Adopt a ministry. Adopt a local government. Adopt a teaching hospital.

2

Put mathematicians in the room where the problem is formulated, not where the answer is being justified.

3

Fix the pipeline. Teach mathematics as a language for describing reality, not as a ritual of memorisation.

4

Build the data. Conduct the census. Digitise the registries. Open the data.

5

Demand quantitative literacy from leadership. Teach them to ask: what is the error bar on that?

6

Bridge to deployment. Mathematics that cannot be deployed is only a beautiful idea.

What we would see.

Health

Drugs arriving before they are needed. Outbreaks flattened by prediction, not by panic.

Education

Teachers posted where the children actually are, not where the posting is convenient.

Agriculture

Post-harvest losses cut by routing and storage. Prices settled by forecasting, not by decree.

Energy

A grid that balances instead of collapsing.

Governance

Budgets that are feasible. Inflated contracts flagged before the money leaves.

Every single item on this list is already a solved problem somewhere in the world today. We are not waiting for a discovery. We are waiting for a decision.

THE FUTURE IS ALREADY HERE

AI is not a computing revolution.

It is a mathematics revolution wearing a computing costume.

Linear algebra

Multivariable
calculus

Probability

Optimisation

Information
theory

Everybody sitting in this hall is already an AI researcher. The subject the whole world is now pouring trillions of dollars into is your own subject. You were doing it before it became fashionable.

EVERY OPPORTUNITY CARRIES ITS OWN DANGER

The other edge of the blade.

Steganography

AI-generated images can now hide payloads that are statistically indistinguishable from ordinary content. A picture is no longer just a picture.

Synthetic media

A voice cloned from thirty seconds of audio. In a country where one false statement can start a fire, this is national security.

Post-quantum

Every classified file and bank record rests on an assumption about factorisation. Other nations are already migrating. Where is our own plan?

Algorithmic bias

A model trained on data that does not include us, now making decisions about us, in credit, in medicine, in visas.

We cannot only consume this mathematics. We must audit it, build it, and defend against it.

SO WHAT DO WE DO FROM HERE?

There is one variable we control.

It is the number of people in this nation who are willing to carry rigour out of the classroom and apply it to the country.

The professor who adopts a ministry.

The student who builds for the local government, not only for the journal.

The young person who makes Nigeria's decisions defensible.

No legislation required. No budget line. No permission. You can start on Monday morning.

A nation is not built by
those who admire the problem.

It is built by those who solve it.

Thank you very much.